

CoRoT-2b

R-0127

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_250725 · created 2026-07-16T05:55:56+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460881.8369 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.022
Transit depth (Rp/Rs)^2	3.33 +/- 0.81 [%]
Orbital Inclination (inc)	83.51 +/- 1.44
Airmass coefficient 1 (a1)	0.948 +/- 0.016
Airmass coefficient 2 (a2)	0.039 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.72 %
Best Comparison Star	#2 - [478, 420]
Optimal Aperture	3.51
Optimal Annulus	9.12
Transit Duration (day)	0.074 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.182 $\pm$ 0.022 vs 0.1667 (deviation 0.7 σ)
Duration fit vs published	0.074 d vs 0.095 d (deviation 1.62 σ)
Mid-time O-C	-6.0 min
Depth SNR	8.3
Residual scatter	1.72 %

Light curve

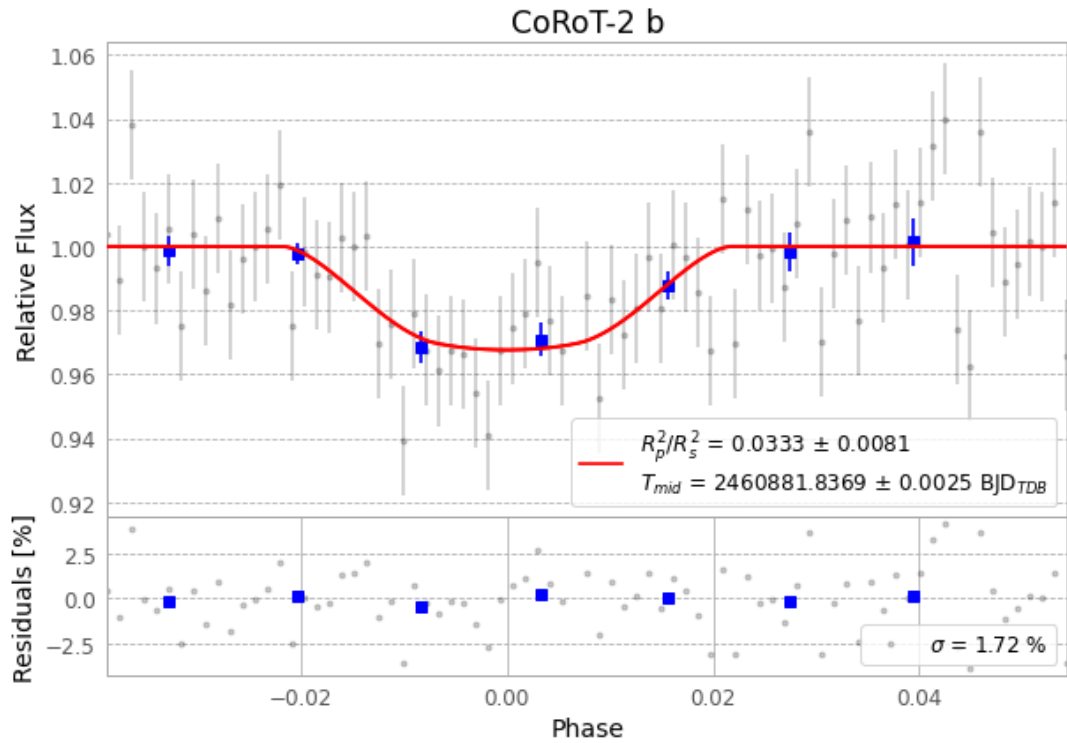


Figure 1 — FinalLightCurve_CoRoT-2 b_2025-07-24.png (SHA-256 4a4e6bc9aca822ad...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [270, 217], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2c06f1cf16c7487f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

79 FITS frames (52 MB), CoRoT-2250725061816.FITS → CoRoT-2250725101217.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-250725.log (70ed5f281297...), FinalLightCurve_CoRoT-2 b_2025-07-24.png (4a4e6bc9aca8...), FinalLightCurve_CoRoT-2 b_2025-07-24.csv (e7987ffb7f21...)

Compute resources

Wall time 145.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 05:55Z.